In [1]:
```
import pandas as pd
dataset= pd.read_csv("salary_dataset.csv")
dataset
```

Out[1]:

	Unnamed: 0	YearsExperience	Salary
0	0	1.2	39344.0
1	1	1.4	46206.0
2	2	1.6	37732.0
3	3	2.1	43526.0
4	4	2.3	39892.0
5	5	3.0	56643.0
6	6	3.1	60151.0
7	7	3.3	54446.0
8	8	3.3	64446.0
9	9	3.8	57190.0
10	10	4.0	63219.0
11	11	4.1	55795.0
12	12	4.1	56958.0
13	13	4.2	57082.0
14	14	4.6	61112.0
15	15	5.0	67939.0
16	16	5.2	66030.0
17	17	5.4	83089.0
18	18	6.0	81364.0
19	19	6.1	93941.0
20	20	6.9	91739.0
21	21	7.2	98274.0
22	22	8.0	101303.0
23	23	8.3	113813.0
24	24	8.8	109432.0
25	25	9.1	105583.0
26	26	9.6	116970.0
27	27	9.7	112636.0
28	28	10.4	122392.0
29	29	10.6	121873.0

In [2]:
```
dataset=dataset.drop(columns=["Unnamed: 0"])
dataset
```

Out[2]:

	YearsExperience	Salary
0	1.2	39344.0
1	1.4	46206.0
2	1.6	37732.0
3	2.1	43526.0
4	2.3	39892.0
5	3.0	56643.0
6	3.1	60151.0
7	3.3	54446.0
8	3.3	64446.0
9	3.8	57190.0
10	4.0	63219.0
11	4.1	55795.0
12	4.1	56958.0
13	4.2	57082.0
14	4.6	61112.0
15	5.0	67939.0
16	5.2	66030.0
17	5.4	83089.0
18	6.0	81364.0
19	6.1	93941.0
20	6.9	91739.0
21	7.2	98274.0
22	8.0	101303.0
23	8.3	113813.0
24	8.8	109432.0
25	9.1	105583.0
26	9.6	116970.0
27	9.7	112636.0
28	10.4	122392.0
29	10.6	121873.0

In [5]:

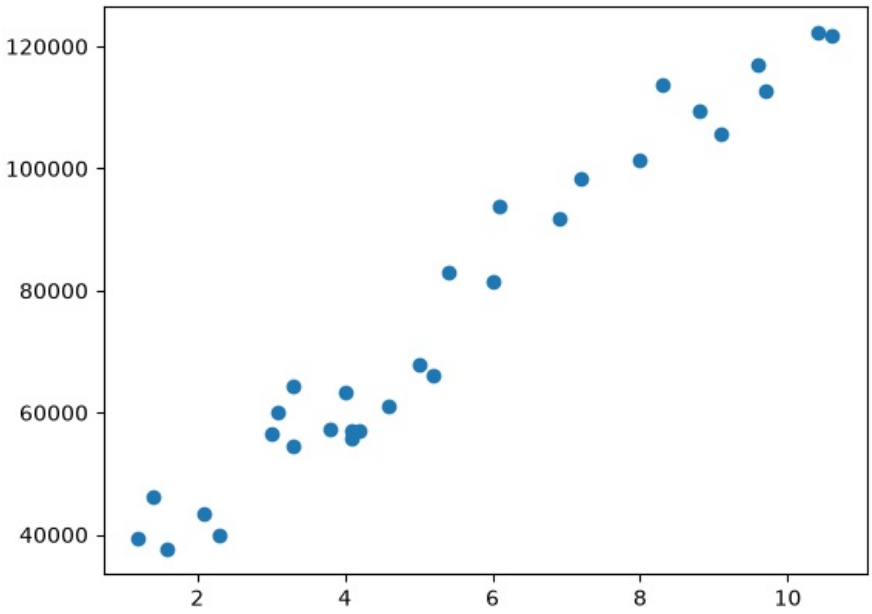
```
x=dataset[["YearsExperience"]]
y=dataset[["Salary"]]
x
y
```

Out[5]:

	Salary
0	39344.0
1	46206.0
2	37732.0
3	43526.0
4	39892.0
5	56643.0
6	60151.0
7	54446.0
8	64446.0
9	57190.0
10	63219.0
11	55795.0
12	56958.0
13	57082.0
14	61112.0
15	67939.0
16	66030.0
17	83089.0
18	81364.0
19	93941.0
20	91739.0
21	98274.0
22	101303.0
23	113813.0
24	109432.0
25	105583.0
26	116970.0
27	112636.0
28	122392.0
29	121873.0

In [6]:

```
import matplotlib.pyplot as plt
plt.scatter(x,y)
plt.show()
```



In [12]:
```
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.20,random_state=42)
y_test
```

Out[12]:

	Salary
27	112636.0
15	67939.0
23	113813.0
17	83089.0
8	64446.0
9	57190.0

In [13]:
```
from sklearn.linear_model import LinearRegression
regerssor=LinearRegression()
regerssor.fit(x_train, y_train)
```

Out[13]:

▾ LinearRegression ⓘ ⓘ

▶ Parameters

▶ Fitted attributes

In [15]:
```
y_pred=regerssor.predict(x_test)
y_pred
```

Out[15]:
```
array([[115791.21011287],
       [ 71499.27809463],
       [102597.86866063],
       [ 75268.80422384],
       [ 55478.79204548],
       [ 60190.69970699]])
```

In [16]:
```
from sklearn.metrics import r2_score
result=r2_score(y_test,y_pred)
result
```

Out[16]:
```
0.9024461774180498
```

In [20]:
```
salary=regerssor.predict([[1]])
salary
```

```
C:\Users\User\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2827: UserWarning: X does not have valid feature names, but LinearRegression was fitted with feature names
  warnings.warn(
```

Out[20]:
```
array([[33804.0168025]])
```

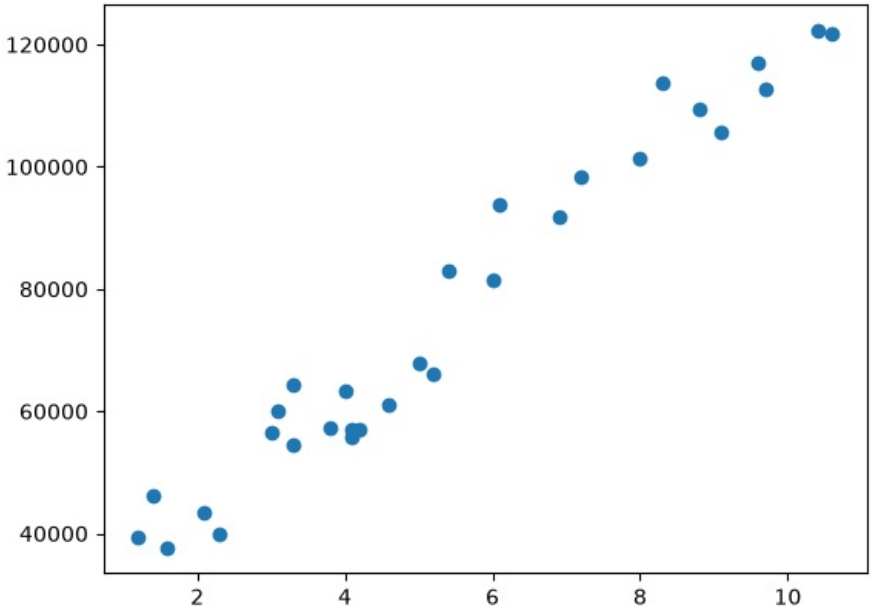
In [21]:
```
m=regerssor.coef_
m
```

Out[21]:
```
array([[9423.81532303]])
```

In [22]:
```
c=regerssor.intercept_
c
```

Out[22]:
```
array([24380.20147947])
```

In [23]:
```
import matplotlib.pyplot as plt
plt.scatter(x,y)
plt.show()
```



In [ ]:

In [ ]: